FitBuddy AI

AI Fitness Plan Generator using Gemini Models

A web-based application for personalized 7-day workout plans, nutrition and recovery tips, feedback-based plan updates, and user management.

Team CIT Hackerz'26

1. PROJECT OVERVIEW

What FitBuddy Does

1. FitBuddy is a web-based AI application that generates personalized workout plans and nutrition tips.
2. It accepts user information such as name, age, weight, fitness goal, and workout intensity.
3. The system generates a structured 7-day fitness plan tailored to the selected goal and intensity.
4. Users can submit feedback and receive an updated plan.
5. SQLite + SQLAlchemy store user information and generated plans.
6. FastAPI provides backend routing and an interactive HTML interface.

2. PROBLEM STATEMENT & OBJECTIVES

Why FitBuddy, and What It Sets Out to Do

Problem/ Need

- Generic workout routines may not match a user's goal or preferred intensity.
- Users may need a simple way to revise a plan when preferences change.
- Nutrition or recovery guidance can be difficult to organize alongside workouts.
- Coaches/admins need a structured view of registered users and plans.

Project Objectives

- Generate goal-specific 7-day workout schedules.
- Provide concise nutrition or recovery tips.
- Use feedback to dynamically revise plans.
- Store original and updated plans for later access.
- Provide both browser UI and API endpoints.
- Support local testing and deployment with FastAPI/Uvicorn.

3. MAJOR FEATURES

Core Functionality Documented for FitBuddy

Personalized Workout Generation

Creates a day-wise 7-day plan.

Nutrition / Recovery Tip

Provides a concise tip aligned with the user's goal.

Feedback-Based Updating

Accepts feedback such as “Add yoga” or “Include more cardio”.

User & Plan Storage

Saves user details, original plan, and updated plan in SQLite.

Admin View

Displays registered users and their original/updated plans.

Web Interface

HTML + Jinja2 pages provide an interactive workflow.

API Testing – FastAPI documentation is available at /docs.

4. TECHNOLOGY STACK

The Tools Behind FitBuddy

Backend & AI

- **Python** — main programming language
- **FastAPI** — web framework and API routing
- **Uvicorn** — ASGI development server
- **Google Gemini** —generative AI layer
- **Pydantic** — input/data validation
- **python-dotenv** — environment variables

Frontend & Data

- **HTML** — user interface structure
- **CSS** — responsive visual styling
- **Jinja2** — dynamic template rendering
- **SQLAlchemy** —ORM/database access
- **SQLite** — persistent local database
- **Git** — version control prerequisite

5. SYSTEM ARCHITECTURE

How a Request Flows Through FitBuddy



Flow: User input → FastAPI route → Gemini generation/update → database storage → Jinja2-rendered result.

6. AI MODEL SELECTION

Model-Selection Details From the Project Documentation

Models Evaluated

- T5 & BART – text generation
- DistilBERT – potential intent classification
- Bloom – general-purpose generation
- Limitations included hosting/local GPU needs and additional fine-tuning or prompting for structured plans.

Selected Gemini Models

- Gemini 1.5 Pro – documented for structured 7-day workout generation and plan updates.
- Gemini Flash – documented for fast nutrition/recovery tips.
- Reasons: API-based integration, structured formatting, low prompt-tuning needs, and fast feedback revisions.

7. APPLICATION WORKFLOW

Six Steps From Input to Stored Plan

1

Enter Details

Name, User ID, age, weight, goal, intensity

2

Generate

FastAPI receives input and calls Gemini

3

View Plan

7-day workout + nutrition/recovery tip

4

Give Feedback

User suggests changes

5

Update

Original plan + feedback sent to Gemini

6

Store

Updated plan saved and displayed

8. BACKEND & API ROUTES

Endpoints That Power the Application

/	Home route: displays the main input form.
/generate-workout	Receives user data, generates workout plan + nutrition tip, saves data, and returns the result.
/submit-feedback	Retrieves the original plan, sends feedback to the AI update function, stores the revised plan, and renders the result.
/view-all-users	Retrieves users and plans for the admin view.

FastAPI Form(...) parameters collect form inputs. **Pydantic** models validate the request structure. Gemini-powered functions are connected from the route handlers.

9. DATABASE & DATA MANAGEMENT

What FitBuddy Persists, and Why

Stored User Data

- User ID
- Name
- Age
- Weight
- Fitness goal
- Workout intensity

Stored Plan Data

- Original AI-generated workout plan
- Updated plan after feedback
- Nutrition/recovery tip
- Records are persisted in SQLite using SQLAlchemy ORM.
- Keeping original and updated versions supports comparison and monitoring.

10. FRONTEND & JINJA2 TEMPLATES

The Pages Behind the User Experience

index.html

Collects user details and submits them to the workout-generation route.

result.html

Displays user information, the AI-generated 7-day plan, nutrition tip, and feedback form.

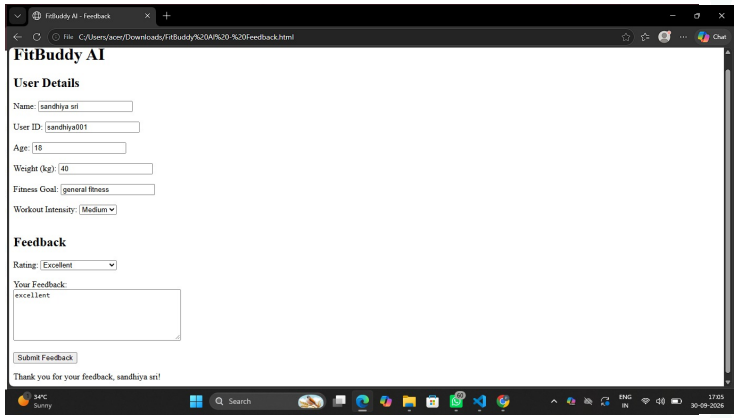
all_users.html

Provides the admin view of users and their plans.

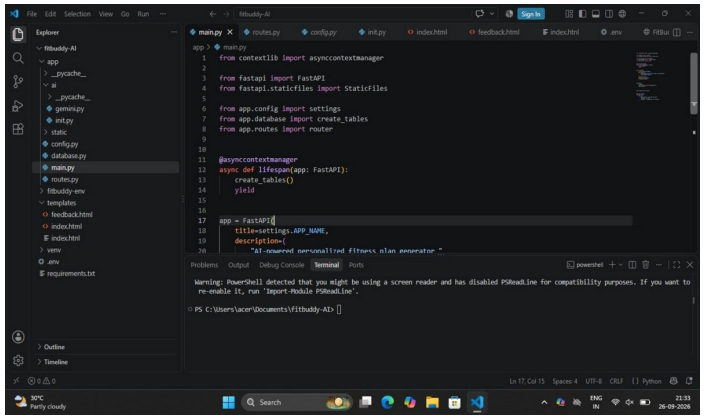
Jinja2 TemplateResponse binds backend values such as username, goal, intensity, workout_plan, and nutrition_tip.
CSS uses Flexbox, responsive media queries, rounded inputs, shadows, hover effects, and a fitness-oriented visual theme.

11. PROJECT SCREENSHOTS

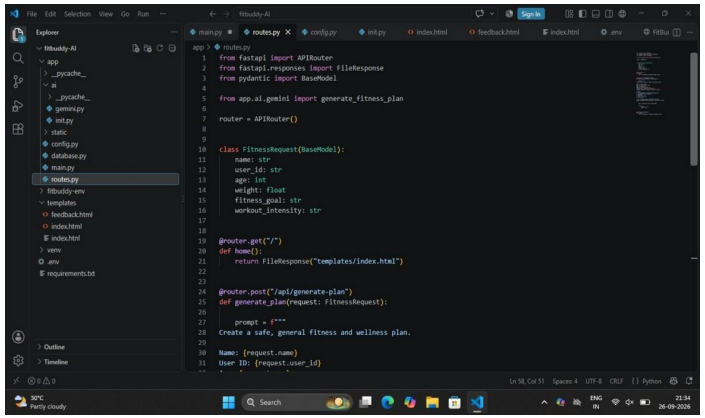
Actual Screens From the Submitted Project



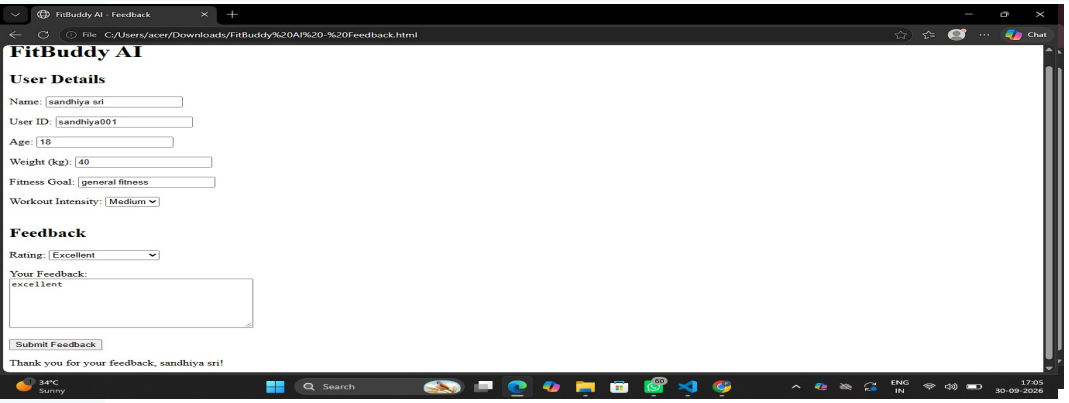
FastAPI main.py



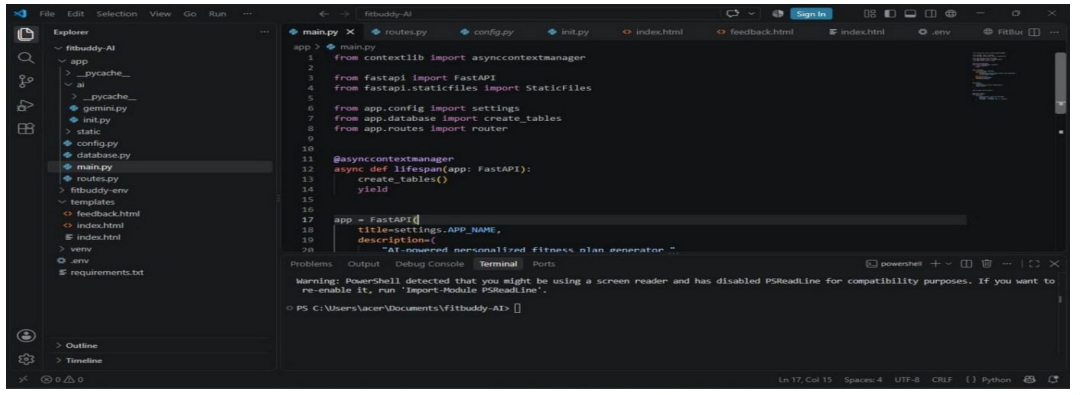
routes.py



main.py configuration



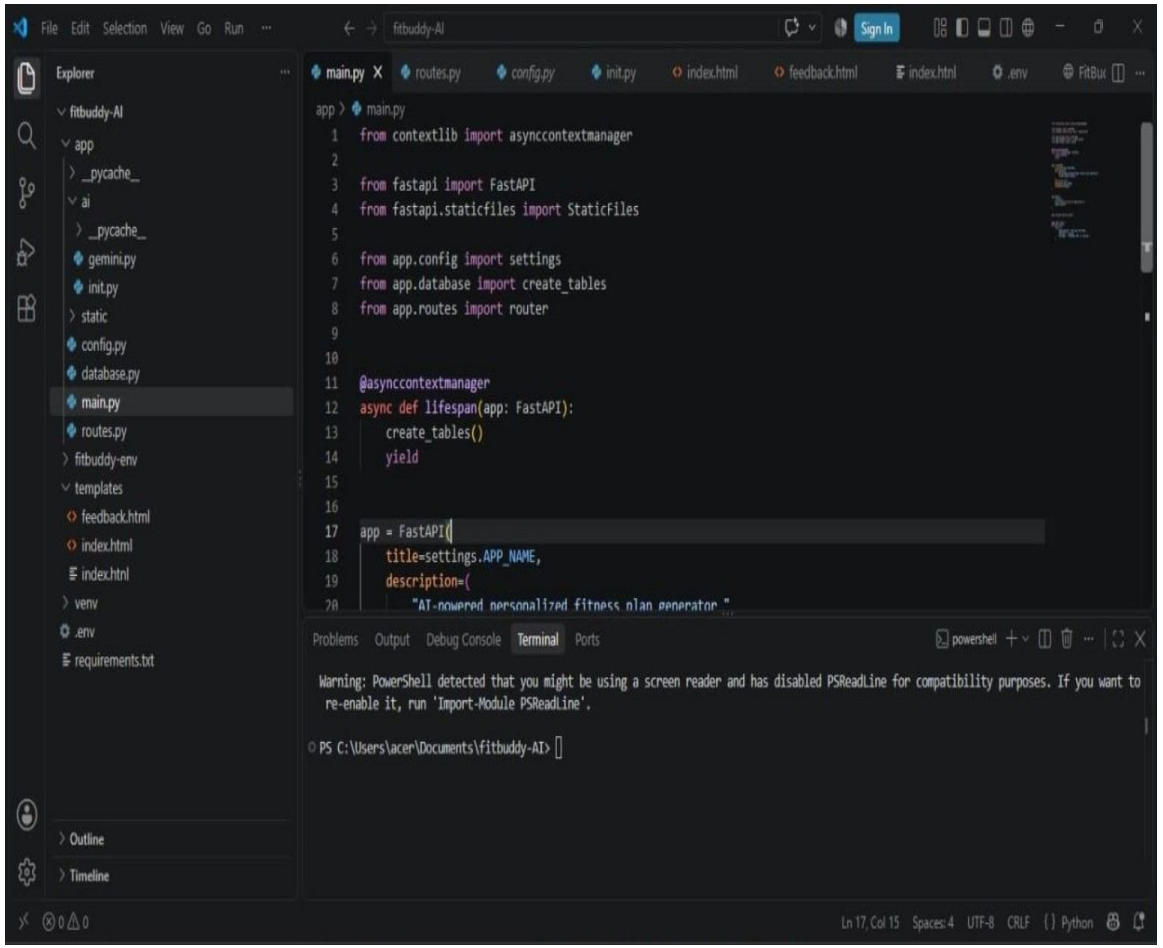
HTML input interface code



Running Feedback page

12. WORKING DEMO — FEEDBACK PAGE

A Closer Look at the Feedback Flow



Running locally at 127.0.0.1:8000/feedback

- User details are displayed on the page.
- Feedback section provides a rating and text box.
- After submission, a confirmation message is shown.
- This demonstrates the feedback stage of the documented workflow.
- The submitted screenshot shows the application running locally at 127.0.0.1:8000/feedback.

13. SETUP & LOCAL DEPLOYMENT

Getting FitBuddy Running on Your Machine

1	**Create a virtual environment**	`python -m venv venv`
2	**Windows activation**	`venv\Scripts\activate`
3	**Install dependencies**	`pip install -r requirements.txt`
4	**Configure the Gemini API key**	Store it in an environment variable or .env file; do not expose the key in source code.
5	**Start the FastAPI server**	`uvicorn app.main:app --reload`
6	**Open the app / view docs**	http://127.0.0.1:8000 • interactive docs at /docs

14. TESTING & VERIFICATION

Confirming FitBuddy Works End to End

Functional Checks

- Home page loads and accepts required user details.
- Workout generation route processes the input.
- AI output is displayed in the result page.
- Nutrition/recovery tip is displayed with the plan.
- Feedback can be submitted to request a revised plan.
- Updated data is stored in the database.

Verification Points

- Run the app locally with Uvicorn.
- Check browser pages through 127.0.0.1:8000.
- Use /docs to inspect/test API endpoints.
- Verify original and updated plans in the admin view.
- Check that environment variables are configured correctly.
- Confirm the database persists records.

15. PROJECT SCENARIOS / USE CASES

How Different Users Experience FitBuddy

Scenario 1 — Personalized Plan

User enters goal and intensity and receives a 7-day plan.

Scenario 2 — Feedback Update

User requests changes such as more cardio or more rest days; the plan is regenerated.

Scenario 3 — Nutrition / Recovery

User receives a concise tip aligned with the selected fitness goal.

Scenario 4 — Admin / Coach

An authorized admin or coach can view users, original plans, and updated plans.

16. FUTURE SCOPE

Where FitBuddy Could Go Next

- Extend the application from local deployment to a cloud deployment.
- Add stronger authentication and role-based access for admin/coach features.
- Add richer progress tracking and visual analytics.
- Expand personalization using additional user preferences and workout history.
- Add more AI-supported wellness features while keeping the existing FastAPI + Gemini architecture.
- Improve UI/UX with richer dashboards and mobile-first components.

17. CONCLUSION

Bringing It All Together

- FitBuddy combines FastAPI, Gemini AI, Jinja2, SQLAlchemy, and SQLite into one web-based fitness planning application.
- It generates personalized 7-day plans, nutrition/recovery tips, and feedback-based updates.
- The application demonstrates an end-to-end flow from user input to AI generation, storage, and web presentation.
- The project also provides an admin view for monitoring original and updated plans.
- Overall, the documented system demonstrates how generative AI can be integrated into a practical wellness-oriented web application.

THANK YOU

FitBuddy AI – Personalized Fitness Plan Generator